

Scoring Rubrics Manual

1. Isolation Rubric

Score	Physical Confinement	Hydrologic Connectivity	Isolation Time (Y)
1	Open water	Free exchange with surroundings	<1 year (open exchange or seasonal turnover)
2	Seasonal or partial cover	Frequent exchange via water/air	1 - 100 years
3	Full perennial cover, thin (<100 m equivalent)	Restricted exchange	100 - 10,000 years
4	Full perennial cover, thick (100 m - 1 km equivalent)	Minimal diffusive exchange or seepage	10,000 - 1,000,000 years
5	Deep overburden (>1 km equivalent, rock or ice)	Effectively sealed	>1,000,000 years

Note:

- Measures isolation from the surface biosphere (microbial contamination), not geological isolation
- The three sub-scores combine as their mean — components are partially substitutable
- Time bins are log-spaced to reflect microbial evolution timescales
- All Enceladus targets score 5 by definition (isolation from Earth's biosphere is total)

2. Redox Rubric

Score	Oxidants/Reductants	Disequilibrium
1	Absent	At or near equilibrium; no exploitable gradient
2	Trace presence	Small gradient; weakly exploitable
3	One species present at meaningful concentration	Moderate gradient; coupling plausible
4	Multiple species at meaningful concentrations	Large gradient; exploitable coupling supported
5	Multiple species at high concentrations (e.g. H ₂ and CH ₄ , CO ₂ and SO ₄ ²⁻)	Sustained flux; Eh below -350 mV or equivalent free energy

Note:

- Scores the chemical state of the site, not how well-characterised it is (characterisation confidence is handled by the reliability rubric)
- Combines as the minimum of the two columns: both must be satisfied for a redox gradient to be bioenergetically exploitable
- "Meaningful concentration" refers to thresholds sufficient for chemolithotrophic growth

3. Reliability Rubric

Score	Source Type	Measurement Type
3	Peer-reviewed primary research	Direct in-situ measurement with calibrated instruments
2	PhD thesis, technical report from established lab, conference proceeding	Indirect measurement, well-validated proxy, or modeling constrained by direct measurements
1	Science journalism, popular press, anecdotal	Unvalidated proxy, distant analogy, or speculation

Note:

- Two independent axes (source venue and measurement type) combine as their maximum — the weaker axis governs.
- Captures source quality only. Disagreement between sources is recorded in the uncertainty range, not here.
- Used as a transparency annotation and filter; does not feed into the suitability score directly.